

Zhongshu Xu

(380) 213-4358 | xu.4202@osu.edu

EDUCATION

The Ohio State University | Ph.D. in Applied Mathematics | GPA: 3.9 Aug 2021 - May 2026

- **Relevant courses:** Probability, Machine Learning, Intermediate Studies in Artificial Intelligence

University of Science and Technology of China | B.S. in Applied Mathematics | GPA: 3.4 Aug 2016 - Jun 2020

- **Relevant courses:** C, Java, Data Structure and Algorithm, Mathematical Statistics

WORKING EXPERIENCE

Risk Program Senior Associate | **JPMorgan Chase & Co.** Jan 2026 – Present

- Develop and deploy **transformer**-based models for fraud detection in large-scale digital payment systems, training on datasets exceeding 10 billion transactions.

INTERN EXPERIENCE

Quantitative Analysis Intern (Model Risk Governance and Review) | **JPMorgan Chase & Co.** Jun 2025 – Aug 2025

- Provide local and global explainability for the signature verification (**ResNet50** + **Siamese**) and handwritten text recognition (**CRNN**) models in Check Fraud Detection pipeline, using black-box and white-box methods to generate saliency heatmaps that verify model focus and highlight letter-level handwriting differences.
- Leveraged explanation results to improve models: (1) identified overly broad attention in ResNet50, enabling removal of the last two stages (94% parameter reduction, **8.5×** speedup, no accuracy loss); (2) replaced **BLSTM** in CRNN with a **Vision Transformer** block to mitigate noisy attention patterns, reducing test set Character Error Rate from **9.9%** to **5.6%**

Machine Learning Intern | **Riverain Technologies** May 2024 – Aug 2024

- Built a robust deep learning pipeline for artery/vein segmentation in pulmonary CT scans using **UNet**-based centerline extraction model + graph-based vessel separation, achieving Dice score >0.98 .
- Enhanced model robustness to real-world artifacts—such as missing data, discontinuities, and overlapping masks—through targeted output space design and synthetic data augmentation.

RESEARCH PROJECTS

Digital Twin for Airfoil Optimization with Reinforcement Learning Feb 2024 – Present

- Developed a **Transformer** model to forecast airfoil surface pressure distributions from historical data, achieving inference speeds **100×** faster than traditional CFD simulations.
- Embedded the model into the **PPO** Algorithms framework as surrogate environment, enabling the agent to dynamically control the airfoil's angle of attack to maximize the lift-to-drag ratio and accelerate aerodynamic optimization.

Learning Stochastic Differential Equations (SDEs) from Data Jan 2023 – Present

- Pioneered Density-Enforced Auto-Encoders (DEAE) for discerning underlying SDEs from time-series data, directing a three-person team in theoretical proofs, model refinement, and rigorous testing.
- Achieved superior performance with our method, which had 5 times smaller errors than established techniques such as Variational Auto-Encoders (VAE) and Generative Adversarial Networks (GANs). Paper published in [JMLMC](#).

Learning Partial Differential Equations (PDEs) from Data Aug 2021 – Present

- Developed a novel spatio-temporal **Transformer** for predicting PDE solution sequences, achieving superior long-term accuracy while using only 10% of the parameters compared to traditional video Transformers.
- Designed a Convolutional Neural Network (**CNN**)-like architecture to efficiently handle PDEs with local operators, reducing the model complexity from $O(N^2)$ to $O(1)$, improving stability and scalability.
- Created a memory-based neural network structure, designed specifically for PDE modeling in scenarios where data is incomplete, expanded the potential applications of neural networks in PDE modeling. Paper published in [JCP](#).

Enhancing CNN for Fourier Operator Learning Jul 2019 - April 2020

- Designed BNet2, a CNN algorithm with convolutional channel connection, which reduced parameters from $O(N^2)$ to $O(N)$ without loss of accuracy.
- Utilized BNet2 with FT Initialization in various applications such as solving PDEs and deblurring / denoising signals. Paper published in [MSML 2020](#).



中国科学技术大学

University of Science and Technology of China

Scholastic Record of University of Science and Technology of China

学业成绩表

Reg.NO: PB16000626	Name: Xu Zhongshu	Gender: Male	Date of Birth: 19980619						
Enrl Date: 20160901	Dep Date:	E.S: 4							
School: Special Class for the Gifted Young									
Major: Information & Computational Science		All Curriculum GPA: 3.4							
Term	Course Title	Gr.	Hrs.	Cr.	Term	Course Title	Gr.	Hrs.	Cr.
2016FA	Chemical Principles B	78	80	4	2016FA	Mechanics	75	80	4
2016FA	Basic Sports	A-	40	1	2016FA	Ideological and Moral Cultivation and Basics of Law	82	60	3
2016FA	Computer Programming A	76	100	4	2016FA	Mathematical Analysis A1	83	120	6
2016FA	Video Assisted Speaking A	93	40	2	2016FA	Intercultural Communication	96	40	2
2016FA	College English	88	80	4	2016FA	Experiment of General Chemistry	B+	40	1
2016FA	Military Theory	Pass	40	1	2017SP	Soccer I	B+	40	1
2017SP	Freshman Seminar	B+	20	1	2017SP	Data Structures and Database	85	90	3.5
2017SP	Fundamentals of Marxism	86	60	3	2017SP	College Physics Experiment I	89	60	1.5
2017SP	Mechanics and Thermal Physics	92	80	4	2017SP	Mathematical Analysis A2	85	120	6
2017SP	Linear Algebra A1	80	100	5					
2017FA	Yoga II	A+	40	1	2017FA	Foundation of Algebra	83	60	3
2017FA	Differential Equations I	88	80	4	2017FA	An Outline of Modern and Contemporary Chinese History	83	40	2
2017FA	Mathematical Analysis A3	85	80	4	2017FA	College Physics Experiment II	88	60	1.5
2017FA	Fundamentals of Java Software Development	B+	40	2	2017FA	Linear Algebra A2	85	80	4
2017FA	Electromagnetism B	93	80	4	2018SP	Optics and Atomic Physics	83	80	4
2018SP	Modern Algebra	88	80	4	2018SP	Introduction to Chinese Important Thoughts	82	60	3
2018SP	Practicum in Ideological and Political Theories	Pass	120	3	2018SP	Complex Analysis	85	80	4
2018SP	Real Analysis	87	80	4	2018SP	Badminton II	B+	40	1
2018FA	Probability Theory	80	80	4	2018FA	Numerical Partial Differential Equations	81	60	3
2018FA	Numerical Algebra	88	60	3	2018FA	Functional Analysis	82	80	4
2018FA	Differential Geometry	76	80	4	2019SP	Differential Equations II	86	80	4
2019SP	Regression Analysis	85	80	4	2019SP	Japanese Culture and Society(Bilingual)	A	40	2
2019SP	Numerical Analysis	85	60	3	2019SP	Mathematical Modeling	A	80	3
2019FA	Mathematical Statistics	69	80	4	2019FA	Mathematical Experiments	B+	40	2
2019FA	Symphony Appreciation	B+	40	2	2019FA	Finite Element Methods	60	80	4
2019FA	Mathematical Software	B	40	2	2019FA	Situation and Policy	Pass	20	1
Graduation Thesis		Name: Dynamic Prediction and Parameter Estimation for Partial Differential Equations based on Neural Networks			Supervisor: Zhang mengning		A	8	2020SP

Note:(H) represents the curriculum of Honors;FA:Fall SP:Spring SU:Summer

GPA Calculation:

Centesimal Grade:100-95 94-90 89-85 84-82 81-78 77-75 74-72 71-68 67-65 64 63-61 60

Letter Grade: A+ A A- B+ B B- C+ C C- D+ D D-

Point Value: 4.3 4 3.7 3.3 3 2.7 2.3 2 1.7 1.5 1.3 1

$$GPA = \frac{\sum(\text{Course Credit} * \text{Course GP})}{\sum \text{Course Credit}}$$



Undergraduate Education Office,
University of Science and Technology of China

Page first/1 Pages

Print date: 2020/6/17



*University of Science and Technology
of China*

awards the

GRADUATE CERTIFICATE

in **Information & Computational Science**

to **Xu Zhongshu**

male, Student No.PB16000626, born on Jun 19, 1998,

having completed a four-year undergraduate program from

Sep 2016 to Jul 2020

Registration No:103581202005000007

Date Issued: Jul 1, 2020

A blue ink signature, likely of the President Bao Xinhe, written in a stylized cursive script.

President Bao Xinhe



中国科学技术大学

毕业证书

徐中恕，男，1998 年 6 月 19
日出生，学号 PB16000626，于 2016
年 9 月至 2020 年 7 月在本校
信息与计算科学 专业四年制本科学
习，修完教学计划规定的全部课程，
成绩合格，准予毕业。



校长

包信和

电子注册号码：

103581202005000007

二〇二〇年七月一日

University of Science and Technology of China

It is hereby certified that
after having satisfied all the requirements of the University
Xu Zhongshu
was on 18 June 2020 admitted to the degree of
Bachelor of Natural Science
in Information and Computing Science



NO.1035842020000271

A stylized, handwritten signature in black ink, appearing to read 'Bao Xin'.

President Bao Xinhe



中国科学技术大学

学士学位证书



徐中恕，男，1998 年 6 月 19 日出生。已完成
信息与计算科学专业 学士学位培养计划。经校学位
评定委员会审议通过，授予 理学 学士学位。

校长 包信和

证书编号：
1035842020000271

二〇二〇年六月十八日

 For: **ZHONGSHU XU**



Most Recent I-94

Note to employers, local, state or federal agency granting benefits:

Please visit the CBP I-94/I-95 Website and click on the tab for "Get Most Recent I-94/I-95" to perform a search for the applicant to confirm that the biographic and travel information displayed on this I-94/I-95 printout matches the "Get Most Recent I-94/I-95" returned results for this applicant. Reference the CBP I-94/I-95 Website FAQs.

Admission I-94 Record Number: 893047284A4

Arrival/Issued Date: 2026 January 08

Class of Admission: F1

Admit Until Date: D/S

Details provided on the I-94 Information form:

Last/Surname: XU

First (Given) Name: ZHONGSHU

Birth Date: 1998 June 19

Document Number: E84744825

Country of Citizenship: China

-
- ▶ Effective April 26, 2013, DHS began automating the admission process. An alien lawfully admitted or paroled into the U.S. is no longer required to be in possession of a preprinted Form I-94/I-95. A record of admission printed from the CBP website constitutes a lawful record of admission. See 8 CFR § 1.4(d).
 - ▶ What to do if someone requests your admission info: If an employer, local, state or federal agency requests admission information, present your admission (I-94/I-95) number along with any additional required documents requested by that employer or agency.
 - ▶ For security, close your browser after retrieving your I-94/I-95 number.

OMB No. 1651-0111
Expiration Date: 02/28/2026

The Ministry of Foreign Affairs of the People's Republic of China requests all civil and military authorities of foreign countries to allow the bearer of this passport to pass freely and afford assistance in case of need.



护照号码/Passport No.

E84744825

徐中恕
XU, ZHONGSHU

出生日期 / Date of birth

19 JUN 1998

北京/BEIJING

11 8月/AUG 2016

北京/BEIJING

10 8月/AUG 2026

公安部出入境管理局
MPS Exit & Entry Administration

持照人签名/Bearer's signature

徐中恕

0839303593

[illegible]

E847448258CHN9806191M2608101NA0MNGNAMLKBA928

08

签证
VISASMOTIVO
DE VIAJE

MEXICO

12 MAR. 2021

MEXICO

02/04/21

02/04/21

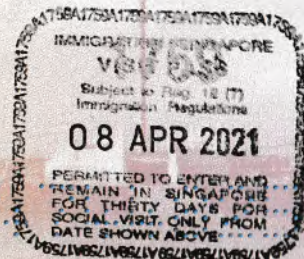
天安門



08

08

09

签证
VISAS

09

09

中国边检 CHINA
2024-12-28
北京【出】
1583

中国边检 CHINA
2021-07-27
深圳机场【出】
0423

U.S. CUSTOMS AND BORDER PROTECTION
ADMITTED
LAX
JUL 27 2021
Class
Unti

Pt
D/S

山西塞上源布

MAY 11 2024

Unit # F1

D/S

N0031426717

ADMITTED
NMC

UNITED STATES CUSTOMS AND BORDER PROTECTION

13

DEC 02 2024
CLASS
UNTIL
F1
DS
N0031420112

VISA

UNITED STATES OF AMERICA



Issuing Post Name
SINGAPORE

Control Number
20211123140050

XU

ZHONGSHU

Passport Number
E84744825

Sex

Birth Date 19JUN1998
Nationality CHIN

Visa Type /Class
R F1

enquiries

28APK2021

Expiration Date
26APR2026

Annotation

N0031420712

THE OHIO STATE UNIVERSITY

P8866224

COLUMBUS, OH

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VNUSAXU<<ZHONGSHU<<<<<<<<<<<<<<<<<<<<<<
E847448258CHN9806191M2604262F1SGPOB48Q572244

15

8101

14250

CANADA

ISSUED AT / DELIVRÉ À DATE OF ISSUE / DATE D'ÉMISSION EXPIRY DATE / DATE D'EXPIRATION

CPC-DITAWA 08/02/2024 09/08/2026
DOCUMENT NO. / N° DU DOCUMENT

MULTIPLE E741702931

V-1 VISITOR

XU, ZHONGSHU

E84744825 ONE

FPC 1132138499

ИЮНЬ 1900

[illegible]

内蒙古 呼伦贝尔草原



Issuing Post Name

BEIJING

Surname

FIX

Given Name

ZHONGSHU

Passport Number

E847

Entries

11

Issue Date

07MAY2019

Sex

三

Birth Date

19 JUN 1998

Expiration Date

30SEP2019

Control Number

20191220460012

Visa Type / Class

R J1

Nationality

CHIN

1000

Annotation

N0030461456; P-1-00242; 01JUL2019-30SEP2019

DUKE UNIV, MEDICAL CENTER, AND HEALTH SYSTEM

SUBJECT TO SECTION 212(E) TWO YEAR RULE

[illegible]

E847448258CHN9806191M1909306J1BEJ37SEX572243

Department of Homeland Security
U.S. Immigration and Customs EnforcementI-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038**SEVIS ID: N0031420712**

SURNAME/PRIMARY NAME Xu	GIVEN NAME Zhongshu	Class of Admission F-1 ACADEMIC AND LANGUAGE
PREFERRED NAME Zhongshu Xu	PASSPORT NAME	
COUNTRY OF BIRTH CHINA	COUNTRY OF CITIZENSHIP CHINA	
CITY OF BIRTH Beijing	DATE OF BIRTH 19 JUNE 1998	
FORM ISSUE REASON CONTINUED ATTENDANCE	ADMISSION NUMBER	

SCHOOL INFORMATION

SCHOOL NAME The Ohio State University The Ohio State University - Columbus	SCHOOL ADDRESS 2009 Millikin Road, 140 Enarson Classroom Building, Columbus, OH 43210
SCHOOL OFFICIAL TO CONTACT UPON ARRIVAL Sarah Homan Student Immigration Coordinator	SCHOOL CODE AND APPROVAL DATE CLE214F00442000 08 JANUARY 2003

PROGRAM OF STUDY

EDUCATION LEVEL DOCTORATE	MAJOR 1 Mathematics, General 27.0101	MAJOR 2 None 00.0000
PROGRAM ENGLISH PROFICIENCY Required	ENGLISH PROFICIENCY NOTES Student is proficient	EARLIEST ADMISSION DATE 25 JULY 2021
START OF CLASSES 24 AUGUST 2021	PROGRAM START/END DATE 24 AUGUST 2021 - 09 MAY 2027	

FINANCIALS

ESTIMATED AVERAGE COSTS FOR: 9 MONTHS		STUDENT'S FUNDING FOR: 9 MONTHS	
Tuition and Fees	\$ 40,507	Personal Funds	\$ 0
Living Expenses	\$ 16,217	GA or Fellowship	\$ 60,781
Expenses of Dependents (0)	\$ 0	Funds From Another Source	\$
Other	\$	On-Campus Employment	\$
TOTAL	\$ 56,724	TOTAL	\$ 60,781

REMARKS**SCHOOL ATTESTATION**

I certify under penalty of perjury that all information provided above was entered before I signed this form and is true and correct. I executed this form in the United States after review and evaluation in the United States by me or other officials of the school of the student's application, transcripts, or other records of courses taken and proof of financial responsibility, which were received at the school prior to the execution of this form. The school has determined that the above named student's qualifications meet all standards for admission to the school and the student will be required to pursue a full program of study as defined by 8 CFR 214.2(f)(6). I am a designated school official of the above named school and am authorized to issue this form.

X	DATE ISSUED	PLACE ISSUED
SIGNATURE OF: Sarah Homan, Student Immigration Coordinator	04 March 2024	Columbus, OH

STUDENT ATTESTATION

I have read and agreed to comply with the terms and conditions of my admission and those of any extension of stay. I certify that all information provided on this form refers specifically to me and is true and correct to the best of my knowledge. I certify that I seek to enter or remain in the United States temporarily, and solely for the purpose of pursuing a full program of study at the school named above. I also authorize the named school to release any information from my records needed by DHS pursuant to 8 CFR 214.3(g) to determine my nonimmigrant status. **Parent or guardian, and student, must sign if student is under 18.**

X	DATE
SIGNATURE OF: Zhongshu Xu	03/05/2024
NAME OF PARENT OR GUARDIAN	DATE
SIGNATURE	ADDRESS (city/state or province/country)
	DATE

Department of Homeland Security
U.S. Immigration and Customs Enforcement

I-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038

SEVIS ID: N0031420712 (F-1)**NAME: Zhongshu Xu****EMPLOYMENT AUTHORIZATIONS**

TYPE	FULL/PART-TIME	STATUS	START DATE	END DATE
CPT	FULL TIME	APPROVED	13 MAY 2024	26 JULY 2024

EMPLOYER INFORMATION

TYPE	AUTHORIZATION DATES			
CPT	13 MAY 2024 - 26 JULY 2024			
EMPLOYER NAME	START DATE	END DATE	CITY & STATE	
Riverain Technologies	13 MAY 2024	26 JULY 2024	MIAMISBURG, OH	

CHANGE OF STATUS/CAP-GAP EXTENSION

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AUTHORIZED REDUCED COURSE LOAD

--

CURRENT SESSION DATES

CURRENT SESSION START DATE	CURRENT SESSION END DATE
08 JANUARY 2024	05 MAY 2024

TRAVEL ENDORSEMENT

This page, when properly endorsed, may be used for re-entry of the student to attend the same school after a temporary absence from the United States. Each endorsement is valid for one year.

Designated School Official	TITLE	SIGNATURE	DATE ISSUED	PLACE ISSUED
Sarah C. Homan	Immigration Coordinator		03/04/2024	Columbus, OH
		X		
		X		
		X		

Department of Homeland Security
U.S. Immigration and Customs Enforcement

I-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038

SEVIS ID: N0031420712

SURNAME/PRIMARY NAME Xu	GIVEN NAME Zhongshu	Class of Admission F-1 ACADEMIC AND LANGUAGE
PREFERRED NAME Zhongshu Xu	PASSPORT NAME	
COUNTRY OF BIRTH CHINA	COUNTRY OF CITIZENSHIP CHINA	
CITY OF BIRTH Beijing	DATE OF BIRTH 19 JUNE 1998	
FORM ISSUE REASON CONTINUED ATTENDANCE	ADMISSION NUMBER	

SCHOOL INFORMATION

SCHOOL NAME The Ohio State University The Ohio State University - Columbus	SCHOOL ADDRESS 2009 Millikin Road, 140 Enarson Classroom Building, Columbus, OH 43210
SCHOOL OFFICIAL TO CONTACT UPON ARRIVAL Lindsey Kinderdine Student Immigration Specialist	SCHOOL CODE AND APPROVAL DATE CLE214F00442000 08 JANUARY 2003

PROGRAM OF STUDY

EDUCATION LEVEL DOCTORATE	MAJOR 1 Mathematics, General 27.0101	MAJOR 2 None 00.0000
PROGRAM ENGLISH PROFICIENCY Required	ENGLISH PROFICIENCY NOTES Student is proficient	EARLIEST ADMISSION DATE 25 JULY 2021
START OF CLASSES 24 AUGUST 2021	PROGRAM START/END DATE 24 AUGUST 2021 - 09 MAY 2027	

FINANCIALS

ESTIMATED AVERAGE COSTS FOR: 9 MONTHS		STUDENT'S FUNDING FOR: 9 MONTHS	
Tuition and Fees	\$ 40,507	Personal Funds	\$ 0
Living Expenses	\$ 16,217	GA or Fellowship	\$ 60,781
Expenses of Dependents (0)	\$ 0	Funds From Another Source	\$
Other	\$	On-Campus Employment	\$
TOTAL	\$ 56,724	TOTAL	\$ 60,781

REMARKS

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SCHOOL ATTESTATION

I certify under penalty of perjury that all information provided above was entered before I signed this form and is true and correct. I executed this form in the United States after review and evaluation in the United States by me or other officials of the school of the student's application, transcripts, or other records of courses taken and proof of financial responsibility, which were received at the school prior to the execution of this form. The school has determined that the above named student's qualifications meet all standards for admission to the school and the student will be required to pursue a full program of study as defined by 8 CFR 214.2(f)(6). I am a designated school official of the above named school and am authorized to issue this form.

☒ **Signed by** Lindsey Kinderdine, Student Immigration Specialist **DATE ISSUED** 13 February 2025 **PLACE ISSUED** Columbus, OH

STUDENT ATTESTATION

I have read and agreed to comply with the terms and conditions of my admission and those of any extension of stay. I certify that all information provided on this form refers specifically to me and is true and correct to the best of my knowledge. I certify that I seek to enter or remain in the United States temporarily, and solely for the purpose of pursuing a full program of study at the school named above. I also authorize the named school to release any information from my records needed by DHS pursuant to 8 CFR 214.3(g) to determine my nonimmigrant status. **Parent or guardian, and student, must sign if student is under 18.**

☒ **Signed by** Zhongshu Xu **DATE** 02/14/2025
SIGNATURE OF: Zhongshu Xu

☒ **Signed by** _____ **DATE** _____
NAME OF PARENT OR GUARDIAN **SIGNATURE** **ADDRESS (city/state or province/country)** **DATE**

Department of Homeland Security
U.S. Immigration and Customs Enforcement

I-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038

SEVIS ID: N0031420712 (F-1)

NAME: Zhongshu Xu

EMPLOYMENT AUTHORIZATIONS

TYPE	FULL/PART-TIME	STATUS	START DATE	END DATE
CPT	FULL TIME	APPROVED	02 JUNE 2025	22 AUGUST 2025

EMPLOYER INFORMATION

TYPE	AUTHORIZATION DATES			
CPT	02 JUNE 2025 - 22 AUGUST 2025			
EMPLOYER NAME	START DATE	END DATE	CITY & STATE	
JPMorgan Chase & Co.	02 JUNE 2025	22 AUGUST 2025	NEW YORK, NY	

CHANGE OF STATUS/CAP-GAP EXTENSION

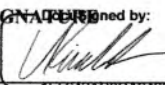
AUTHORIZED REDUCED COURSE LOAD

CURRENT SESSION DATES

CURRENT SESSION START DATE	CURRENT SESSION END DATE
06 JANUARY 2025	04 MAY 2025

TRAVEL ENDORSEMENT

This page, when properly endorsed, may be used for re-entry of the student to attend the same school after a temporary absence from the United States. Each endorsement is valid for one year.

Designated School Official	TITLE	SIGNATURE	DATE ISSUED	PLACE ISSUED
Lindsey Kinderdine	Specialist	X  1F5642AC8CA640C...	02/13/2025	Columbus, OH
		X		
		X		
		X		

Department of Homeland Security
U.S. Immigration and Customs Enforcement

I-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038

SEVIS ID: N0031420712

SURNAME/PRIMARY NAME
Xu

PREFERRED NAME
Zhongshu Xu

COUNTRY OF BIRTH
CHINA

CITY OF BIRTH
Beijing

FORM ISSUE REASON
CONTINUED ATTENDANCE

GIVEN NAME
Zhongshu

PASSPORT NAME

COUNTRY OF CITIZENSHIP
CHINA

DATE OF BIRTH
19 JUNE 1998

ADMISSION NUMBER

Class of Admission

F-1

ACADEMIC AND
LANGUAGE

SCHOOL INFORMATION

SCHOOL NAME
The Ohio State University
The Ohio State University - Columbus

SCHOOL OFFICIAL TO CONTACT UPON ARRIVAL
Devan Gibson
Student Immigration Specialist

SCHOOL ADDRESS
2009 Millikin Road, 140 Enarson Classroom Building,
Columbus, OH 43210

SCHOOL CODE AND APPROVAL DATE
CLE214F00442000
08 JANUARY 2003

PROGRAM OF STUDY

EDUCATION LEVEL
DOCTORATE

MAJOR 1
Mathematics, General 27.0101

MAJOR 2
None 00.0000

PROGRAM ENGLISH PROFICIENCY
Required

ENGLISH PROFICIENCY NOTES
Student is proficient

EARLIEST ADMISSION DATE
25 JULY 2021

START OF CLASSES
24 AUGUST 2021

PROGRAM START/END DATE
24 AUGUST 2021 - 09 MAY 2027

FINANCIALS

ESTIMATED AVERAGE COSTS FOR: 9 MONTHS

Tuition and Fees	\$	40,507
Living Expenses	\$	16,217
Expenses of Dependents (0)	\$	0
Other	\$	

TOTAL	\$	56,724
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STUDENT'S FUNDING FOR: 9 MONTHS

Personal Funds	\$	0
GA or Fellowship	\$	60,781
Funds From Another Source	\$	
On-Campus Employment	\$	

TOTAL	\$	60,781
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REMARKS

SCHOOL ATTESTATION

I certify under penalty of perjury that all information provided above was entered before I signed this form and is true and correct. I executed this form in the United States after review and evaluation in the United States by me or other officials of the school of the student's application, transcripts, or other records of courses taken and proof of financial responsibility, which were received at the school prior to the execution of this form. The school has determined that the above named student's qualifications meet the standards for admission to the school and the student will be required to pursue a full program of study as defined by 8 CFR 214.2(f)(6). I am a designated school official of the above named school and am authorized to issue this form.

☒ Devan Gibson
SIGNATURE OF: Devan Gibson, Student Immigration
Specialist

DATE ISSUED
10 November 2025

PLACE ISSUED
Columbus, OH

STUDENT ATTESTATION

I have read and agreed to comply with the terms and conditions of my admission and those of any extension of stay. I certify that all information provided on this form refers specifically to me and is true and correct to the best of my knowledge. I certify that I seek to enter or remain in the United States temporarily, and solely for the purpose of pursuing a full program of study at the school named above. I also authorize the named school to release any information from my records needed by DHS pursuant to 8 CFR 214.3(g) to determine my nonimmigrant status. **Parent or guardian, and student, must sign if student is under 18.**

☒ Zhongshu Xu
SIGNATURE OF: Zhongshu Xu

Nov 12 2025
DATE

NAME OF PARENT OR GUARDIAN

☒ SIGNATURE

ADDRESS (city/state or province/country)

DATE

Department of Homeland Security
U.S. Immigration and Customs Enforcement

I-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038

SEVIS ID: N0031420712 (F-1)

NAME: Zhongshu Xu

EMPLOYMENT AUTHORIZATIONS

TYPE	FULL/PART-TIME	STATUS	START DATE	END DATE
CPT	FULL TIME	APPROVED	26 JANUARY 2026	10 MAY 2026

EMPLOYER INFORMATION

TYPE	AUTHORIZATION DATES			
CPT	26 JANUARY 2026 - 10 MAY 2026			
EMPLOYER NAME	START DATE	END DATE	CITY & STATE	
JPMorganChase	26 JANUARY 2026	10 MAY 2026	COLUMBUS, OH	

CHANGE OF STATUS/CAP-GAP EXTENSION

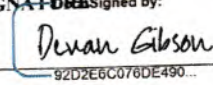
AUTHORIZED REDUCED COURSE LOAD

CURRENT SESSION DATES

CURRENT SESSION START DATE	CURRENT SESSION END DATE
26 AUGUST 2025	21 DECEMBER 2025

TRAVEL ENDORSEMENT

This page, when properly endorsed, may be used for re-entry of the student to attend the same school after a temporary absence from the United States. Each endorsement is valid for one year.

Designated School Official	TITLE	SIGNATURE	DATE ISSUED	PLACE ISSUED
Devan Gibson	Immigration Specialist	 92D2E6C078DE490...	11/10/2025	Columbus, OH
		X		
		X		
		X		
		X		

Department of Homeland Security
U.S. Immigration and Customs Enforcement

I-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038

SEVIS ID: N0031420712

SURNAME/PRIMARY NAME Xu	GIVEN NAME Zhongshu	Class of Admission F-1 ACADEMIC AND LANGUAGE
PREFERRED NAME Zhongshu Xu	PASSPORT NAME	
COUNTRY OF BIRTH CHINA	COUNTRY OF CITIZENSHIP CHINA	
CITY OF BIRTH Beijing	DATE OF BIRTH 19 JUNE 1998	
FORM ISSUE REASON CONTINUED ATTENDANCE	ADMISSION NUMBER 893047284A4	

SCHOOL INFORMATION

SCHOOL NAME The Ohio State University The Ohio State University - Columbus	SCHOOL ADDRESS 2009 Millikin Road, 140 Enarson Classroom Building, Columbus, OH 43210
SCHOOL OFFICIAL TO CONTACT UPON ARRIVAL Sooyeon Lee Senior Student Immigration Specialist	SCHOOL CODE AND APPROVAL DATE CLE214F00442000 08 JANUARY 2003

PROGRAM OF STUDY

EDUCATION LEVEL DOCTORATE	MAJOR 1 Mathematics, General 27.0101	MAJOR 2 None 00.0000
PROGRAM ENGLISH PROFICIENCY Required	ENGLISH PROFICIENCY NOTES Student is proficient	EARLIEST ADMISSION DATE 25 JULY 2021
START OF CLASSES 24 AUGUST 2021	PROGRAM START/END DATE 24 AUGUST 2021 - 10 MAY 2026	

FINANCIALS

ESTIMATED AVERAGE COSTS FOR: 9 MONTHS		STUDENT'S FUNDING FOR: 9 MONTHS	
Tuition and Fees	\$ 40,507	Personal Funds	\$ 0
Living Expenses	\$ 16,217	GA or Fellowship	\$ 60,781
Expenses of Dependents (0)	\$ 0	Funds From Another Source	\$
Other	\$	On-Campus Employment	\$
TOTAL	\$ 56,724	TOTAL	\$ 60,781

REMARKS

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SCHOOL ATTESTATION

I certify under penalty of perjury that all information provided above was entered before I signed this form and is true and correct. I executed this form in the United States after review and evaluation in the United States by me or other officials of the school of the student's application, transcripts, or other records of courses taken and proof of financial responsibility, which were received at the school prior to the execution of this form. The school has determined that the above named student's qualifications meet all standards for admission to the school and the student will be required to pursue a full program of study as defined by 8 CFR 214.2(f)(6). I am a designated school official of the above named school and am authorized to issue this form.

SIGNATURE OF: Sooyeon Lee, Senior Student Immigration Specialist	DATE ISSUED 11 February 2026	PLACE ISSUED Columbus, OH
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STUDENT ATTESTATION

I have read and agreed to comply with the terms and conditions of my admission and those of any extension of stay. I certify that all information provided on this form refers specifically to me and is true and correct to the best of my knowledge. I certify that I seek to enter or remain in the United States temporarily, and solely for the purpose of pursuing a full program of study at the school named above. I also authorize the named school to release any information from my records needed by DHS pursuant to 8 CFR 214.3(g) to determine my nonimmigrant status. **Parent or guardian, and student, must sign if student is under 18.**

SIGNATURE OF: Zhongshu Xu	DATE 02/11/2026
NAME OF PARENT OR GUARDIAN	DATE
SIGNATURE	ADDRESS (city/state or province/country)
DATE	

Department of Homeland Security
U.S. Immigration and Customs Enforcement

I-20, Certificate of Eligibility for Nonimmigrant Student Status
OMB NO. 1653-0038

SEVIS ID: N0031420712 (F-1)

NAME: Zhongshu Xu

EMPLOYMENT AUTHORIZATIONS

TYPE	FULL/PART-TIME	STATUS	START DATE	END DATE
CPT	FULL TIME	APPROVED	26 JANUARY 2026	10 MAY 2026
POST-COMPLETION OPT	FULL TIME	REQUESTED	11 MAY 2026	10 MAY 2027

EMPLOYER INFORMATION

TYPE	AUTHORIZATION DATES			
CPT	26 JANUARY 2026 - 10 MAY 2026			
EMPLOYER NAME	START DATE	END DATE	CITY & STATE	
JPMorganChase	26 JANUARY 2026	10 MAY 2026	COLUMBUS, OH	

CHANGE OF STATUS/CAP-GAP EXTENSION

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AUTHORIZED REDUCED COURSE LOAD

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CURRENT SESSION DATES

CURRENT SESSION START DATE	CURRENT SESSION END DATE
12 JANUARY 2026	10 MAY 2026

TRAVEL ENDORSEMENT

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Designated School Official	TITLE	SIGNATURE	DATE ISSUED	PLACE ISSUED
Sooyeon Lee	Sr. Immigration Specialist	X	02/11/2026	Boston, MA
		X		
		X		
		X		



View Application To Graduate

Zhongshu Xu, xu.4202, 500502374

STUDENT INFORMATION **FORM RESPONSES** HISTORY COMM **Approved**

Form Responses

Graduation Information

Graduation Term: Spring 2026

Graduation Level: Doctoral

Graduation Plan: MATH-PH - Mathematics

End of Semester Option: No

Commencement

Attendance: I will attend commencement

Committee

Advisor:

OSUID: 300006458

Name .# xiu.16

First Name: Dongbin

Last Name: Xiu

Middle Name:

Faculty Email: xiu.16@osu.edu

Faculty Type: Current

Phone:

Program: MATH

Category: P

Committee Members:

Faculty Member: Yulong Xing
(xing.205) - MATH - Category P -
Current

Faculty Member: Maria Han Veiga
(hanveiga.1) - MATH - Category P -
Current

External Members:

First Name: -

Last Name: -

Email Address: -

Co-Advisor (if applicable)

OSUID: -

Name .# -

First Name: -

Last Name: -

Middle Name: -

Faculty Email: -

Faculty Type: -

Phone: -

Program: -

Category: -

Commencement Attendance

- ☒ I will attend commencement
- ☐ I will not attend commencement, I will pick up my diploma within one week after commencement
- ☐ I will not attend commencement, I am authorizing someone else to pick up my diploma
- ☐ I will not attend commencement, please mail my diploma to the permanent address below

Save

Cancel Application



THE OHIO STATE UNIVERSITY

Graduate School

250 University Hall
230 N. Oval Mall
Columbus, OH 43210

Phone: [614-292-6031](tel:614-292-6031)Fax: [614-292-3656](tel:614-292-3656)Graduation Services: Grad-SchoolGraduationServices@osu.eduRegistration Services: Grad-SchoolRegistrationServices@osu.edu



View Application For Final Examination

Zhongshu Xu, xu.4202, 500502374

STUDENT INFORMATION **FORM RESPONSES** HISTORY CC Initial Review

Form Responses

Graduation Information

Graduation Term: Spring 2026	Final Examination Type: First Final Exam
Graduation Level: Doctoral	I would like to request a Graduate Faculty representative (Required for Second Final Exam).
Graduation Plan: MATH-PH - Mathematics	No
	Format Review Complete: No

Draft Dissertation/D.M.A.

Draft Dissertation Title: Flow Map Learning for Partial Differential Equations

Oral portion of final examination

There will be a video conference component to this examination. No

Date of Oral Examination:

Date: 04/15/2026
Start Time: 10:00 AM
End Time: 12:00 PM

Room Number: 105
Building: Mathematics Building

Committee

Category M Approved: No

Advisor:

OSUID: 300006458
Name .# xiu.16

External Members:

First Name: -
Last Name: -

First Name: Dongbin
Last Name: Xiu
Middle Name:
Faculty Email: xiu.16@osu.edu
Faculty Type: Current
Phone:
Program: MATH
Category: P

Email Address: -

Graduate Faculty Representative:

OSUID: -
Name .# -
First Name: -
Last Name: -
Middle Name: -
Faculty Email: -
Faculty Type: -
Phone: -
Program: -
Category: -

Co-Advisor (if applicable)

OSUID: -
Name .# -
First Name: -
Last Name: -
Middle Name: -
Faculty Email: -
Faculty Type: -
Phone: -
Program: -
Category: -

Committee Members:

Faculty Member: Yulong Xing
(xing.205) - MATH - Category P -
Current

Faculty Member: Maria Han Veiga
(hanveiga.1) - MATH - Category P -
Current



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